Vacuum Moduli Spaces of Four-Dimensional $\mathcal{N} = 2$ SCFTs

A self-contained guide from protected operators to low-energy geometry

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Abstract

These notes explain how the vacuum branches of a four-dimensional $\mathcal{N} = 2$ superconformal field theory arise from protected local operators and how their geometry controls, but does not by itself exhaust, the low-energy physics. No Lagrangian presentation is assumed.

The discussion deliberately separates three layers that are often conflated: the protected operator ring, the algebraic geometry of its vacuum characters, and the additional dynamical data of the low-energy theory. This separation clarifies why a physical vacuum is a closed point rather than the whole spectrum, why $\operatorname{Spec}\mathcal{R}_{\mathrm{CB}}$ alone does not produce photons or Seiberg–Witten periods, and why a free tangent sector need not be the complete infrared theory.

The intended use is both linear reading and later lookup. Definitions state the precise content; intuition and reasoning boxes record why the definitions are natural and which extra assumptions enter each implication.

Contents

Reader's Guide

The central question

Given the local operator data of an $\mathcal{N} = 2$ SCFT, which protected expectation values label supersymmetric vacua, what geometry do those vacua form, and what massless theory describes fluctuations around a chosen point?

The answer has three logically distinct layers:

Layer	Basic object	What it determines
Operator algebra	a protected ring $\mathcal{R}$	allowed holomorphic order parameters and their relations
Vacuum geometry	characters $\mathcal{R} \to \mathbb{C}$ and $\operatorname{Spec} \mathcal{R}$	ordinary vacua, local rings, tangent directions, and singularities
Low-energy dynamics	massless multiplets, charge lattices, and effective couplings	metrics, central charges, duality monodromies, and the strict IR fixed point

How to use the note

- For a first pass, read sections 2 to 4 and 6 and the final reasoning map in section 8.
- To recover notation quickly, use the conceptual dictionary in section 7.
- For Coulomb special geometry, start at sections 3.3, 3.4 and 3.6.
- For the $\widehat{\mathcal{B}}_R$ notation and flavor moment maps, start at sections 4.1 and 4.3.

The two implications that must never be silently reversed

$$\text{physical vacuum} \Longrightarrow \text{character of the protected ring}$$

does not, without an additional completeness assertion, prove that every algebraic character is physically realized. Likewise,

$$\text{branch tangent modes become free in the strict IR}$$

does not prove that the complete infrared theory contains no residual sector.

Throughout, the theory is assumed unitary and four-dimensional, and "flavor symmetry" means a continuous internal symmetry commuting with the full $\mathcal{N} = 2$ superconformal algebra. Signs and normalizations of $U(1)_r$ charge and of the central charge vary in the literature; invariant statements are emphasized whenever possible.

1 What Data Constitute a Four-Dimensional $\mathcal{N} = 2$ SCFT?

Purpose of this section

This section fixes the minimum common language used later: states, local operators, OPE data, symmetries, and superconformal multiplets. Readers already comfortable with radial quantization and $\mathfrak{su}(2, 2|2)$ may skip directly to section 2.

1.1 A preliminary caveat

There is no single completely rigorous mathematical definition that presently covers every interacting four-dimensional quantum field theory. For the present purpose, it is therefore useful to describe a QFT by the physical data from which its observables and consistency conditions are determined.

Definition 1.1 (Quantum field theory data)**.** A local quantum field theory on four-dimensional Minkowski space is characterized operationally by the following mutually compatible structures:

(i) a space of quantum states;
(ii) a collection of local operators;
(iii) correlation functions of these operators;
(iv) an operator product expansion encoding short-distance products;
(v) a unitary action of spacetime and internal symmetries;
(vi) locality, unitarity, and positivity conditions;
(vii) a prescription for selecting a vacuum and constructing excitations above it.

The individual terms in this definition are explained below.

1.2 States and the Hilbert space

Definition 1.2 (Hilbert space)**.** The *Hilbert space* $\mathcal{H}_{\text{phys}}$ is the complex vector space of physical quantum states equipped with a positive-definite inner product

$$\langle \psi \mid \chi \rangle.$$

A normalized state satisfies $\langle \psi \mid \psi \rangle = 1$.

The inner product determines probabilities. If a state is expanded in an orthonormal basis as

$$|\psi\rangle = \sum_n c_n |n\rangle,$$

then $|c_n|^2$ is the probability of obtaining the outcome associated with $|n\rangle$.

Definition 1.3 (Observable)**.** An *observable* is represented by a self-adjoint operator acting on $\mathcal{H}_{\text{phys}}$. Its expectation value in a normalized state $|\psi\rangle$ is

$$\langle A \rangle_\psi = \langle \psi | A | \psi \rangle.$$

Definition 1.4 (Unitarity)**.** A quantum theory is *unitary* if time evolution preserves the Hilbert-space inner product. If $U(t)$ is the time-evolution operator, then

$$U(t)^\dagger U(t) = \mathrm{id}.$$

Unitarity ensures conservation of total probability and requires physical states to have nonnegative norm.

Remark 1.5. In relativistic QFT, the most basic objects are usually local operator-valued distributions rather than ordinary bounded operators. Their products must be interpreted inside correlation functions or after smearing with test functions.

1.3 Local operators and locality

Definition 1.6 (Local operator)**.** A *local operator* $\mathcal{O}(x)$ is an operator inserted at a spacetime point x. It probes the theory in an arbitrarily small neighborhood of x.

Examples in a Lagrangian theory include a scalar field $\phi(x)$, the field strength $F_{\mu\nu}(x)$, and gauge-invariant composite operators such as $\mathrm{Tr}\,\phi^k(x)$.

Definition 1.7 (Locality)**.** A relativistic QFT is *local* if observables inserted at spacelike-separated points cannot influence one another. For bosonic operators this is expressed by

$$[\mathcal{O}_1(x), \mathcal{O}_2(y)] = 0, \qquad (x-y)^2 > 0,$$

with an analogous graded commutator for fermionic operators.

Definition 1.8 (Lorentz scalar)**.** A *Lorentz scalar operator* is unchanged by Lorentz transformations apart from the transformation of its insertion point. It carries Lorentz spins

$$(j_1, j_2) = (0, 0).$$

A constant nonzero VEV can preserve Lorentz invariance only when it belongs to a scalar operator.

Locality is what permits one to speak about independent measurements in causally disconnected regions.

1.4 Correlation functions

Definition 1.9 (Correlation function)**.** Given a vacuum $|\Omega\rangle$, an n-point correlation function is

$$\langle \mathcal{O}_1(x_1)\cdots\mathcal{O}_n(x_n)\rangle := \langle\Omega|\mathcal{O}_1(x_1)\cdots\mathcal{O}_n(x_n)|\Omega\rangle,$$

with an appropriate time-ordering or Euclidean prescription.

Correlation functions contain the measurable dynamical information of the theory. Their singularities describe short-distance propagation, while their long-distance behavior describes the spectrum of low-energy states.

Definition 1.10 (Connected correlator)**.** A *connected correlator* is the part of a correlation function that does not factorize into products of correlators involving disjoint subsets of insertions. For example,

$$\langle \mathcal{O}_1\mathcal{O}_2\rangle_{\mathrm{conn}} = \langle \mathcal{O}_1\mathcal{O}_2\rangle - \langle \mathcal{O}_1\rangle\langle \mathcal{O}_2\rangle\,.$$

Definition 1.11 (Cluster decomposition)**.** Let p be a pure vacuum, meaning a vacuum in a single superselection sector rather than a coherent superposition or statistical mixture of distinct vacua. The vacuum satisfies *cluster decomposition* if, for local operators at large spacelike separation,

$$\lim_{|\mathbf{x}-\mathbf{y}|\to\infty} \langle \mathcal{O}_1(t,\mathbf{x})\mathcal{O}_2(t,\mathbf{y})\rangle_p = \langle \mathcal{O}_1\rangle_p \langle \mathcal{O}_2\rangle_p .$$

Equivalently,

$$\lim_{|\mathbf{x}-\mathbf{y}|\to\infty} \langle \mathcal{O}_1(t,\mathbf{x})\mathcal{O}_2(t,\mathbf{y})\rangle_{p,\mathrm{conn}} = 0.$$

Cluster decomposition expresses the statistical independence of experiments separated by an arbitrarily large distance within one pure vacuum. It is a long-distance statement; by itself it says nothing about the coincident-point product of two operators.

1.5 The operator product expansion

Definition 1.12 (Operator product expansion)**.** The *operator product expansion*, or OPE, expresses the product of two nearby local operators as a sum of local operators:

$$\mathcal{O}_i(x)\mathcal{O}_j(0) \sim \sum_k C_{ij}{}^k(x)\,\mathcal{O}_k(0), \qquad x \to 0.$$

The functions $C_{ij}{}^k(x)$ are called *OPE coefficients* or *Wilson coefficients*.

The symbol $\sim$ means equality inside correlation functions as an asymptotic expansion at short distance. In a conformal field theory, the coordinate dependence of $C_{ij}{}^k(x)$ is strongly constrained, and the remaining numerical coefficients are part of the conformal data.

Remark 1.13. Associativity of the OPE means that a product of several operators gives the same result regardless of the order in which pairwise OPEs are performed. In four-point functions, this becomes the crossing-symmetry constraint.

1.6 Symmetries, charges, and conserved currents

Definition 1.14 (Symmetry)**.** A *symmetry* is a transformation acting on states and operators that leaves all physical predictions invariant. Infinitesimally, a continuous symmetry is generated by a conserved charge Q.

Definition 1.15 (Spacetime and internal symmetries)**.** A *spacetime symmetry* acts on spacetime coordinates and therefore on the placement of operators. An *internal symmetry* acts on internal labels of states and operators while leaving spacetime points fixed.

Definition 1.16 (Global symmetry and gauge redundancy)**.** A *global symmetry* acts nontrivially on physical states and gauge-invariant operators. A *gauge symmetry* is instead a redundancy of a particular description: gauge-related field configurations represent the same physical configuration. Consequently, local physical operators must be gauge invariant.

For an operator $\mathcal{O}$, the infinitesimal action of the symmetry is represented schematically by

$$\delta\mathcal{O} = i[Q,\mathcal{O}].$$

Definition 1.17 (Conserved current)**.** A *conserved current* is a local vector operator $J^\mu(x)$ obeying

$$\partial_\mu J^\mu = 0.$$

The corresponding conserved charge is

$$Q = \int_{\mathbb{R}^3} d^3x \, J^0(x).$$

Definition 1.18 (Stress-energy tensor)**.** The *stress-energy tensor* $T_{\mu\nu}$ is the conserved current associated with spacetime translations. It obeys

$$\partial^\mu T_{\mu\nu} = 0.$$

Its conserved charges are the four-momentum operators P_ν.

Definition 1.19 (Flavor symmetry)**.** A *flavor symmetry* is a global internal symmetry that commutes with the entire superconformal algebra. Its group will be denoted by F. Flavor transformations act nontrivially on local operators but do not rotate the supercharges.

This distinguishes flavor symmetry from R-symmetry, which acts nontrivially on the supercharges themselves.

Definition 1.20 (Poincaré symmetry)**.** *Poincaré symmetry* is the symmetry generated by translations P_μ and Lorentz transformations $M_{\mu\nu}$. It is the spacetime symmetry of a relativistic theory on flat Minkowski space.

Definition 1.21 (Conformal field theory)**.** A *conformal field theory*, or CFT, is a local QFT invariant under the conformal group, which extends Poincaré transformations by dilatations and special conformal transformations.

Definition 1.22 (Supersymmetric field theory)**.** A *supersymmetric field theory* has fermionic symmetry generators called supercharges. Their anticommutators contain spacetime translations, so supersymmetry relates bosonic and fermionic operators and states.

Definition 1.23 (Superconformal field theory)**.** A *superconformal field theory*, or SCFT, is simultaneously conformal and supersymmetric, with both sets of generators combined into a superconformal algebra.

1.7 Conformal symmetry

The bosonic conformal algebra in four dimensions is $\mathfrak{so}(4,2)$. Its generators are

$$P_\mu, \qquad M_{\mu\nu}, \qquad D, \qquad K_\mu,$$

representing translations, Lorentz transformations, dilatations, and special conformal transformations.

Definition 1.24 (Scaling dimension)**.** A local operator $\mathcal{O}(0)$ has *scaling dimension* Δ if

$$[D, \mathcal{O}(0)] = i\Delta\mathcal{O}(0)$$

in Lorentzian conventions. Equivalently, under a scale transformation $x \mapsto \lambda x$,

$$\mathcal{O}(x) \mapsto \lambda^{-\Delta}\mathcal{O}(\lambda x).$$

Definition 1.25 (Conformal primary and descendant)**.** A *conformal primary operator* is annihilated at the origin by all special conformal generators:

$$[K_\mu, \mathcal{O}(0)] = 0.$$

Operators obtained by acting with translations P_μ are called *conformal descendants*.

1.8 Supersymmetry and the $\mathcal{N} = 2$ superconformal algebra

Four-dimensional $\mathcal{N} = 2$ Poincaré supersymmetry contains eight real supercharges, conventionally written as

$$Q^I_\alpha, \qquad \widetilde{Q}_{I\dot{\alpha}}, \qquad I = 1, 2.$$

Here α and $\dot{\alpha}$ are left- and right-handed Lorentz spinor indices, while I is an $SU(2)_R$ index.

The full $4d$ $\mathcal{N} = 2$ superconformal algebra is $\mathfrak{su}(2, 2|2)$. Its bosonic subalgebra is

$$\mathfrak{so}(4, 2) \oplus \mathfrak{su}(2)_R \oplus \mathfrak{u}(1)_r.$$

The last two factors form the R-symmetry algebra.

The notation and shortening terminology used below follow the systematic multiplet classification of Córdova, Dumitrescu, and Intriligator [1].

Definition 1.26 (R-symmetry)**.** An *R-symmetry* is an internal symmetry that acts nontrivially on the supercharges. For $4d$ $\mathcal{N} = 2$ theories, the connected R-symmetry group is locally

$$SU(2)_R \times U(1)_r.$$

We denote the $SU(2)_R$ spin of an operator by

$$R = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots.$$

The corresponding irreducible representation has dimension $2R + 1$. The normalization and sign of the $U(1)_r$ charge differ across the literature, so Coulomb-branch operators will primarily be labeled by their scaling dimensions Δ.

Definition 1.27 (Superconformal primary)**.** A *superconformal primary* is annihilated at the origin by the special conformal generators K_μ and by the conformal supercharges S and $\widetilde{S}$.

Definition 1.28 (Superconformal multiplet)**.** A *superconformal multiplet* is the collection of operators generated from one superconformal primary by repeated action of the Poincaré supercharges $Q, \widetilde{Q}$ and translations P_μ.

Definition 1.29 (Long and short multiplets)**.** A *long multiplet* has no additional null relations beyond those required by the algebra. A *short multiplet* satisfies extra annihilation conditions because its quantum numbers saturate a unitarity bound. Such shortened multiplets are often called *BPS multiplets*.

Definition 1.30 (Protected scalar operator)**.** In these notes, a *protected scalar operator* means a specified Lorentz-scalar superconformal primary that obeys specified supercharge-annihilation conditions and whose scaling dimension is fixed by the corresponding shortening relation. The annihilating supercharges and the other quantum numbers are part of the definition; they differ between the Coulomb and Higgs sectors.

Definition 1.31 (Protected operator ring)**.** A *protected operator ring* $\mathcal{R}$ is a unital complex vector space of protected scalar operators, or their protected-cohomology classes, equipped with a product

$$\mathcal{O}_1 \cdot \mathcal{O}_2 \in \mathcal{R}$$

obtained from the regular protected term in their OPE. The product is required to close in the chosen protected sector and to be associative and commutative after the relevant cohomological or contact-term identifications.

The phrase "BPS operator" will only be used as shorthand for such a specified protected primary. A BPS *multiplet* contains the primary together with many supercharge descendants, and those descendants are not all scalar order parameters or moduli-space coordinates. Thus "this operator generates a BPS multiplet" and "every operator in that multiplet is a branch coordinate" are very different statements.

BPS shortening protects certain operator dimensions and permits the intrinsic identification of Coulomb- and Higgs-branch primaries even when no Lagrangian is known. Their precise shortening conditions will be given before their rings are defined.

1.9 The operator spectrum and conformal data

Definition 1.32 (Operator spectrum)**.** The *operator spectrum* is the collection of local operators, organized into irreducible superconformal multiplets and labeled by their quantum numbers.

For a superconformal primary, the relevant labels include

$$(\Delta; j_1, j_2; R; r; \rho_F),$$

where:

- Δ is the scaling dimension;
- (j_1, j_2) are the spins under the Lorentz group $SU(2)_1 \times SU(2)_2$;
- R is the $SU(2)_R$ spin;
- r is the $U(1)_r$ charge;
- ρ_F is a representation of the flavor group F.

Definition 1.33 (Conformal data)**.** The *conformal data* consist of the spectrum of conformal primary operators together with their OPE coefficients. In a unitary CFT, these data are constrained by unitarity, locality, OPE associativity, and crossing symmetry.

In this operational sense, specifying all consistent conformal data specifies the local observables of the CFT.

1.10 State–operator correspondence

Definition 1.34 (State–operator correspondence)**.** In a CFT, radial quantization identifies a local operator inserted at the origin with a state on the three-sphere:

$$\mathcal{O}(0)|0\rangle \longleftrightarrow |\mathcal{O}\rangle \in \mathcal{H}_{S^3}.$$

The scaling dimension of $\mathcal{O}$ equals the energy of the corresponding state on S^3.

This correspondence allows statements about local operators to be translated into statements about states and superconformal representations.

2 Vacua and Vacuum Moduli Spaces

From a state to a point of geometry

The key task is to justify the chain

$$\begin{array}{rcl} \text{supersymmetric vacuum} & \longrightarrow & \text{VEVs of protected operators} \\ & & \Downarrow \\ \text{character of a protected ring} & \longrightarrow & \text{closed point of an affine spectrum.} \end{array}$$

The Ward-identity and cluster-decomposition steps in the middle are essential; multiplicativity is not automatic for arbitrary local operators.

2.1 Vacuum states

Definition 2.1 (Poincaré-invariant vacuum)**.** A *Poincaré-invariant vacuum* is a state $|\Omega\rangle$ satisfying

$$P_\mu|\Omega\rangle = 0, \qquad M_{\mu\nu}|\Omega\rangle = 0.$$

It is translationally and Lorentz invariant.

Definition 2.2 ($\mathcal{N} = 2$ supersymmetric vacuum)**.** A Poincaré-invariant vacuum is an $\mathcal{N} = 2$ *supersymmetric vacuum* if it is annihilated by all eight real Poincaré supercharges:

$$Q^I_\alpha|\Omega\rangle = 0, \qquad \widetilde{Q}_{I\dot\alpha}|\Omega\rangle = 0.$$

Definition 2.3 (Superconformal vacuum)**.** A *superconformal vacuum* is invariant under the full superconformal group. In particular, it is invariant under dilatations, special conformal transformations, and conformal supercharges in addition to Poincaré supersymmetry.

For an SCFT with a moduli space, the superconformal vacuum is the distinguished point at which all positive-dimension order parameters vanish. It is usually called the *origin* or the *tip of the cone*.

2.2 Vacuum expectation values

Definition 2.4 (Vacuum expectation value)**.** The *vacuum expectation value*, or VEV, of a local operator $\mathcal{O}$ in a vacuum p is

$$\langle\mathcal{O}\rangle_p := \langle\Omega_p|\mathcal{O}|\Omega_p\rangle.$$

Translation invariance implies that the VEV of a local operator is independent of its insertion point. Lorentz invariance implies that a nonzero VEV can be assigned only to a Lorentz scalar, unless Lorentz symmetry is spontaneously broken.

Remark 2.5. The statement that a vacuum preserves supersymmetry is not identical to the statement that every operator with a nonzero VEV is annihilated by every supercharge. Rather, the vacuum state itself is annihilated by all Poincaré supercharges, while the operators whose VEVs parameterize supersymmetric vacua lie in suitable short multiplets.

2.3 The vacuum moduli space

Definition 2.6 (Vacuum moduli space)**.** The *vacuum moduli space* $\mathcal{M}_{\text{vac}}$ is the space of inequivalent Poincaré-invariant vacua that minimize the energy. The $\mathcal{N} = 2$ supersymmetric vacuum moduli space consists of those vacua that preserve all eight Poincaré supercharges.

The word *modulus* means a continuously variable parameter labeling inequivalent vacua. A *branch* is a connected or irreducible component, or more generally a geometrically distinguished stratum, of the vacuum moduli space.

Definition 2.7 (Order parameter)**.** An *order parameter* is an operator whose VEV distinguishes different vacua. On a supersymmetric moduli space, specified protected scalar primaries can serve as holomorphic order parameters.

Suppose a set of scalar operators O_i obeys algebraic relations

$$f_a(O_1, \ldots, O_n) = 0.$$

Their VEVs

$$u_i = \langle O_i \rangle$$

must satisfy

$$f_a(u_1, \ldots, u_n) = 0.$$

Thus operator relations become equations defining the vacuum geometry.

For the branch chiral rings used below, supersymmetric Ward identities imply a further property. Schematically, translations of a protected insertion are exact with respect to a supercharge that annihilates both the insertions and the supersymmetric vacuum,

$$\partial_\mu O(x) = \{Q, \Lambda_\mu(x)\}.$$

Consequently, separated correlators of compatible protected insertions are independent of their separation. This is the missing bridge between the coincident protected OPE product and the long-distance statement of cluster decomposition. It is not true for arbitrary QFT operators.

Proposition 2.8 (Vacua as characters of an operator ring)**.** *Let $\mathcal{R}$ be a protected operator ring for which the preceding Ward-identity position independence holds, and let p be a normalized pure supersymmetric vacuum satisfying cluster decomposition. Then*

$$\chi_p : \mathcal{R} \longrightarrow \mathbb{C}, \qquad \chi_p(O) = \langle O \rangle_p \,.$$

is a unital $\mathbb{C}$-algebra homomorphism. In particular,

$$\chi_p(O_1 \cdot O_2) = \chi_p(O_1)\chi_p(O_2).$$

Proof. Linearity of expectation values gives $\mathbb{C}$-linearity, and normalization gives

$$\chi_p(1) = 1.$$

For two protected operators, the definition of the protected OPE product and the Ward identity allow the insertions to be separated without changing the correlator. Therefore

$$\begin{aligned}\chi_p(O_1 \cdot O_2) &= \left\langle O_1(x)O_2(y)\right\rangle_p \\ &= \lim_{|\mathbf{x}-\mathbf{y}|\to\infty} \left\langle O_1(x)O_2(y)\right\rangle_p \\ &= \langle O_1\rangle_p \langle O_2\rangle_p ,\end{aligned}$$

where the final equality is cluster decomposition. □

The kernel

$$\mathfrak{m}_p := \ker \chi_p$$

is a maximal ideal, since the unital $\mathbb{C}$-algebra map has image $\mathbb{C}$ and hence

$$\mathcal{R}/\mathfrak{m}_p \simeq \mathbb{C}.$$

Thus a complex-valued physical vacuum determines a closed $\mathbb{C}$-point, not an arbitrary prime ideal.

Remark 2.9. This proposition proves the direction

$$\text{physical vacuum} \longrightarrow \text{character of the protected ring.}$$

The converse claim that every algebraic character is realized by a physical vacuum is an additional physical assertion implicit in identifying the complete branch with the ring spectrum.

Why the VEV map is multiplicative

$$\begin{aligned}\chi_p(O_1O_2) &= \langle O_1(x)O_2(y)\rangle_p && \text{protected OPE and position independence,} \\ &= \lim_{|x-y|\to\infty} \langle O_1(x)O_2(y)\rangle_p && \text{move the insertions apart,} \\ &= \langle O_1\rangle_p\langle O_2\rangle_p && \text{cluster decomposition.}\end{aligned}$$

The first equality is the supersymmetric input; the last is the vacuum-sector input.

2.4 Ordinary vacua, maximal ideals, and the affine spectrum

Definition 2.10 (Affine spectrum and complex points)**.** For a commutative ring R,

$$\operatorname{Spec} R = \{\mathfrak{p} \subset R : \mathfrak{p} \text{ is prime}\}, \qquad \operatorname{MaxSpec} R = \{\mathfrak{m} \subset R : \mathfrak{m} \text{ is maximal}\}.$$

If R is a $\mathbb{C}$-algebra, its ordinary complex points are

$$(\operatorname{Spec} R)(\mathbb{C}) := \operatorname{Hom}_{\mathbb{C}\text{-alg}}(R, \mathbb{C}).$$

For a finitely generated $\mathbb{C}$-algebra, Hilbert's Nullstellensatz identifies these complex points with $\operatorname{MaxSpec} R$.

Accordingly, a single vacuum is neither the whole branch nor the whole spectrum. It corresponds to one maximal ideal $\mathfrak{m}_p$. The notation $\operatorname{Spec} R$ packages the ordinary closed points together with the algebraic functions and their local behavior.

For example,

$$\operatorname{Spec} \mathbb{C}[u]$$

contains the closed points $(u - a)$, $a \in \mathbb{C}$, as well as the non-closed prime (0). The latter is the generic point of the whole irreducible affine line; it is not an extra vacuum with a definite complex value of u.

The most useful extra structure is often the local ring

$$\mathcal{O}_{\operatorname{Spec} R,p} = R_{\mathfrak{m}_p},$$

which records holomorphic functions and relations near the vacuum. It determines, for example, the Zariski tangent space

$$T_p \operatorname{Spec} R \simeq \operatorname{Hom}_{\mathbb{C}}\left(\mathfrak{m}_p/\mathfrak{m}_p^2, \mathbb{C}\right)$$

and helps distinguish smooth from singular points.

The scheme can also remember nilpotent information invisible to ordinary complex points. The rings

$$\mathbb{C}[u] \qquad \text{and} \qquad \mathbb{C}[u, \varepsilon]/(\varepsilon^2)$$

have the same complex-valued points, because every character sends ε to zero, but different local rings and tangent spaces. Whether such a nonreduced structure carries physical information must be decided theory by theory; many familiar Coulomb chiral rings are assumed to be reduced, and often freely generated.

Thus, if one only wants the set of ordinary complex vacua, $\operatorname{MaxSpec} R$ or $\operatorname{Hom}_{\mathbb{C}\text{-alg}}(R, \mathbb{C})$ is enough. The standard notation $\operatorname{Spec} R$ is used when one also wants generic subvarieties, local rings, multiplicities, infinitesimal structure, and functorial operations such as quotients and localization.

Why keep $\operatorname{Spec} R$?

The non-closed points are not extra physical vacua. They are bookkeeping devices for generic behavior on irreducible subvarieties. The genuinely useful gain is the attached local algebra: it remembers which functions exist near a vacuum, how branches meet, whether the point is smooth, and whether infinitesimal or nilpotent structure is present.

2.5 Spontaneous breaking of conformal symmetry

Let $\mathcal{O}$ be a scalar operator of positive scaling dimension Δ, and suppose

$$\langle \mathcal{O} \rangle = v \neq 0.$$

The VEV defines a physical scale

$$M_v \sim |v|^{1/\Delta}.$$

A nonzero scale is incompatible with an invariant conformal vacuum. Therefore, a generic nonzero VEV spontaneously breaks

$$D, \qquad K_\mu, \qquad S, \qquad \widetilde{S},$$

while it may preserve

$$P_\mu, \qquad M_{\mu\nu}, \qquad Q, \qquad \widetilde{Q}.$$

Thus one should distinguish

the SCFT as a theory   from   a particular vacuum of that theory.

The theory possesses superconformal symmetry, but a nonzero-VEV vacuum generally preserves only $\mathcal{N} = 2$ Poincaré supersymmetry.

3 The Coulomb Branch

What is intrinsic and what is additional

The Coulomb chiral ring intrinsically defines an affine complex geometry. The $U(1)^r$ photons, electromagnetic lattice, central charge, metric, and Seiberg–Witten periods arise only after adding the low-energy $\mathcal{N} = 2$ dynamics on the physical regular locus.

3.1 Coulomb-branch operators

Definition 3.1 (Coulomb-branch chiral primary)**.** A *Coulomb-branch chiral primary* is a superconformal primary $\mathcal{O}(0)$ satisfying

$$j_1 = j_2 = 0, \qquad R = 0,$$

and annihilated by all four Poincaré supercharges of one chirality, which we conventionally take to be

$$\widetilde{Q}_{I\dot{\alpha}}\mathcal{O}(0) = 0, \qquad I = 1, 2, \quad \dot{\alpha} = 1, 2.$$

The shortening relation fixes its scaling dimension in terms of its nonzero $U(1)_r$ charge, with a sign and normalization that depend on convention.

The short multiplet generated from such a primary is conventionally called an $\mathcal{E}$-type or $\overline{\mathcal{E}}$-type multiplet, depending on which chirality and $U(1)_r$ convention is used [1]. In these notes, "$\mathcal{E}$-type" is only a classification label for the explicit conditions above; it is not an additional ingredient in the definition.

Remark 3.2. The Coulomb-branch coordinate is the bottom scalar primary $\mathcal{O}$, not the entire $\mathcal{E}$-type multiplet. Its fermionic and higher descendants are indispensable members of the supersymmetry representation but are not independent Lorentz-invariant vacuum coordinates.

Primary versus multiplet

A multiplet is the supersymmetry orbit generated from one primary. A vacuum coordinate must itself be a bosonic Lorentz scalar with a constant VEV. The primary has precisely this role; most descendants have spin, fermion parity, or derivative structure that prevents them from being independent Lorentz-invariant order parameters.

Proposition 3.3 (Flavor neutrality of Coulomb primaries)**.** *Every Coulomb-branch chiral primary is a singlet under every continuous flavor symmetry.*

Proof. Let O_A transform in a putative flavor representation,

$$[F^a, O_A] = (T^a)_A{}^B O_B.$$

Choose an $\mathcal{N} = 1$ subalgebra contained in the $\mathcal{N} = 2$ algebra. The Coulomb primary is chiral for this subalgebra. The $\mathcal{N} = 1$ flavor Ward identity says that a nonzero matrix T^a requires the scalar bottom component of the flavor-current multiplet to appear in the $O_A \times \overline{O}^B$ OPE with coefficient proportional to $(T^a)_A{}^B$.

In an $\mathcal{N} = 2$ SCFT, that scalar belongs to the moment-map triplet $\mu^{a(ij)}$, the superconformal primary of the $\widehat{\mathcal{B}}_1$ flavor-current multiplet described below. But O_A and $\overline{O}^B$ are both $SU(2)_R$ singlets, whereas $\mu^{a(ij)}$ has $R = 1$. Hence

$$\langle O_A \overline{O}^B \mu^{a(ij)} \rangle = 0$$

by $SU(2)_R$ selection rules. The Ward identity therefore forces $T^a = 0$. □

This selection-rule viewpoint also underlies mixed Coulomb/moment-map correlator analyses such as [3].

The chirality assumption is essential in this argument: an arbitrary long-multiplet operator with $R = 0$ need not be flavor neutral. Continuous flavor transformations consequently do not move points of the pure Coulomb base. Flavor data can nevertheless affect Coulomb-branch physics through flavor-charged BPS states and background mass parameters in Z_γ.

The protected OPE product of two Coulomb-branch chiral primaries again has a Coulomb-branch chiral primary as its protected component, modulo protected-cohomology and operator relations.

Definition 3.4 (Coulomb-branch chiral ring)**.** The *Coulomb-branch chiral ring* $\mathcal{R}_{\mathrm{CB}}$ is the commutative ring generated by Coulomb-branch chiral primaries, modulo their operator relations.

Let u_i denote their VEVs:

$$u_i = \langle O_i \rangle\,.$$

Definition 3.5 (Coulomb branch)**.** The *Coulomb branch* is a family of supersymmetric Poincaré-invariant vacua. It is parameterized by VEVs of Coulomb-branch chiral primaries, with Higgs-type order parameters set to zero. Its affine algebraic avatar is

$$C = \operatorname{Spec} \mathcal{R}_{\mathrm{CB}}.$$

The ordinary complex-valued Coulomb vacua are the closed points

$$C(\mathbb{C}) = \mathrm{Hom}_{\mathbb{C}\text{-alg}}(\mathcal{R}_{\mathrm{CB}}, \mathbb{C}) \simeq \operatorname{MaxSpec} \mathcal{R}_{\mathrm{CB}},$$

under the usual finite-generation assumption. A vacuum p corresponds to

$$\mathfrak{m}_p = \ker\Big(O \longmapsto \langle O \rangle_p\Big)\,.$$

The entire Coulomb branch is not one maximal ideal. Moreover, $\operatorname{Spec} \mathcal{R}_{\mathrm{CB}}$ encodes only the affine complex algebraic or scheme structure. It does not by itself imply the existence of photons, a metric, an electromagnetic charge lattice, or Seiberg–Witten periods; those arise from additional low-energy $\mathcal{N} = 2$ dynamics.

3.2 Basic geometric properties

Proposition 3.6. *The Coulomb branch is a complex cone with its superconformal vacuum at the tip. If u_i has scaling dimension Δ_i, the complexified scaling action is locally*

$$u_i \longmapsto \lambda^{\Delta_i} u_i, \qquad \lambda \in \mathbb{C}^*.$$

Definition 3.7 (Rank)**.** The *rank* of a four-dimensional $\mathcal{N} = 2$ theory is the complex dimension of the Coulomb branch at a generic smooth point:

$$\mathrm{rk}(\mathcal{T}) = \dim_{\mathbb{C}} C.$$

This is the local dimension of the branch, not necessarily the number of displayed chiral-ring generators: generators can obey relations.

Definition 3.8 (Physical regular locus)**.** The *physical regular locus* $C^\circ \subset C$ is the open region on which the charged low-energy sector has a nonsingular Abelian description, the vector-multiplet kinetic matrix is nondegenerate, and no additional electrically or magnetically charged state is massless. Its complement

$$\Delta_{\text{phys}} := C \setminus C^\circ$$

is the physical discriminant locus.

The physical regular locus must not be identified blindly with the scheme-theoretic smooth locus. A point of the base variety can be algebraically smooth while the special Kähler metric and Abelian effective action are singular because an extra charged state becomes massless.

3.3 Why rank r gives r Abelian vector multiplets

The relevant massless 4d $\mathcal{N} = 2$ Poincaré multiplets have different scalar content:

	bosonic fields	scalar moduli
vector multiplet	A_μ and one complex scalar φ	1 complex direction
hypermultiplet	4 real scalars	1 quaternionic direction

A hypermultiplet contains no photon.

Why does a tangent direction to a vacuum branch give a massless scalar? In a Lagrangian description, let $\Phi^A(t)$ be a curve of degenerate minima of the scalar potential. At a vacuum Φ_0 its tangent v^A obeys

$$v^A \left.\frac{\partial^2 V}{\partial \Phi^A \partial \Phi^B}\right|_{\Phi_0} v^B = 0.$$

The Hessian is the scalar mass-squared matrix, so the tangent fluctuation is massless. Intrinsically, the same conclusion follows by allowing a slowly varying vacuum parameter m^i to become a field $m^i(x)$: the effective action begins with derivative terms and has no potential along the moduli space.

The implication "modulus $\Rightarrow$ massless scalar" does not by itself decide which supersymmetry multiplet contains that scalar. That is extra $\mathcal{N} = 2$ representation-theoretic information. Coulomb-type directions, identified in the UV by the $\mathcal{E}$-type shortening condition, become vector-multiplet scalar directions in the infrared; Higgs-type directions, identified by $\widehat{\mathcal{B}}_R$ shortening, become hypermultiplet directions.

Proposition 3.9 (Generic Coulomb multiplets)**.** *Let $p \in C^\circ$ lie on a rank-r Coulomb branch. The low-energy theory contains exactly r massless Abelian vector multiplets associated with the Coulomb base. Consequently its local gauge algebra contains*

$$\mathfrak{u}(1)^r.$$

Indeed, T_pC contains r independent complex Coulomb moduli, while one $\mathcal{N} = 2$ vector multiplet contains exactly one complex scalar. Each of the resulting r massless spin-one fields requires an Abelian gauge redundancy. At a physical regular point there are no extra massless charged vector bosons that would complete these photons into a non-Abelian algebra. Conversely, an additional massless vector multiplet would contain another freely variable complex scalar and would increase the local Coulomb rank.

This statement is a structural input from the low-energy realization of $\mathcal{N} = 2$ supersymmetry; it does *not* follow from the algebraic identity $C = \operatorname{Spec} \mathcal{R}_{\mathrm{CB}}$ alone.

Why the number is exactly r

$$\dim_{\mathbb{C}} T_pC = r \Longrightarrow r \text{ massless complex Coulomb moduli} \Longrightarrow r \text{ vector multiplets} \Longrightarrow \mathfrak{u}(1)^r$$

at a physical regular point. Hypermultiplets may coexist, but they do not alter this count. An additional vector multiplet would bring an additional complex scalar modulus and hence increase the rank.

Remark 3.10 (Two different $U(1)$s)**.** The UV superconformal $U(1)_r$ is a global R-symmetry that rotates supercharges and Coulomb primaries. The $U(1)^r_{\mathrm{gauge}}$ factors in the generic infrared theory are gauge redundancies with photons. A Coulomb primary's $U(1)_r$ charge helps identify the type of order parameter; it does not generate or equal an infrared electric gauge charge. In fact, a nonzero Coulomb VEV generally breaks $U(1)_r$, while the scalar in an Abelian vector multiplet is neutral under its own Abelian gauge group.

The r vector multiplets need not exhaust every massless field. There can also be h neutral hypermultiplets over a generic Coulomb point, giving locally an enhanced Coulomb or mixed branch with a hyperkähler fiber. Charged hypermultiplets are instead generically massive and become massless only on special loci. Thus r counts vector multiplets associated with the Coulomb base, not all massless multiplets.

3.4 The $\mathcal{N} = 2$ Poincaré central charge

A nonzero Coulomb VEV breaks conformal symmetry but preserves $\mathcal{N} = 2$ Poincaré supersymmetry. The latter algebra admits a complex central operator $\widehat{Z}$. Up to convention-dependent constants and the placement of complex conjugation,

$$\{Q^I_\alpha, \widetilde{Q}_{J\dot\beta}\} = 2\delta^I{}_J \sigma^\mu_{\alpha\dot\beta} P_\mu,$$

and

$$\{Q^I_\alpha, Q^J_\beta\} = 2\epsilon_{\alpha\beta}\epsilon^{IJ}\,\overline{\widehat{Z}}, \qquad \{\widetilde{Q}_{I\dot\alpha}, \widetilde{Q}_{J\dot\beta}\} = 2\epsilon_{\dot\alpha\dot\beta}\epsilon_{IJ}\,\widehat{Z}.$$

It is central with respect to the super-Poincaré generators. It is unrelated to the Virasoro central charge of a two-dimensional CFT and to the four-dimensional Weyl-anomaly coefficients usually denoted a and c.

Central extensions of supersymmetry and their relation to BPS bounds were established in [4]; their exact realization on a four-dimensional $\mathcal{N} = 2$ Coulomb branch is central to the Seiberg–Witten solution [5,6].

Fix a Coulomb vacuum p and a superselection sector of asymptotic states. Schur's lemma says only that $\widehat{Z}$ acts by a complex number on each irreducible super-Poincaré representation:

$$\widehat{Z}|\Psi\rangle = z|\Psi\rangle.$$

It does not, by itself, say what z is or that it is determined by electromagnetic charge.

The stronger statement comes from the concrete QFT realization of the supersymmetry algebra. The supercharges are spatial integrals of the supercurrent. Their anticommutator contains a total derivative, so the central operator is a surface charge at spatial infinity [4,5]:

$$\widehat{Z} = \int_{S^2_\infty} (\text{scalar fields})\,(\text{electric and magnetic fluxes}).$$

In a local Abelian frame this gives the operator identity, up to normalization,

$$\widehat{Z} = \sum_{I=1}^{r} \left(a^I(p)\widehat{n}_e^I + a_{D,I}(p)\widehat{n}_{m,I}\right) + \sum_A m^A \widehat{s}_A.$$

Here $\widehat{n}_e^I$ and $\widehat{n}_{m,I}$ measure electric and magnetic fluxes, while $\widehat{s}_A$ are flavor charges and m^A are background flavor-vector-multiplet masses. The flavor term is absent when the masses vanish.

The vacuum p enters because it fixes the scalar boundary conditions at infinity,

$$\varphi^I(x) \longrightarrow a^I(p) \qquad (|x| \to \infty),$$

and the charge sector enters because it fixes the flux eigenvalues. Thus a state with full charge label

$$\gamma = (n_e^I, n_{m,I}, s_A)$$

has

$$Z_\gamma(p) = \sum_I \left(n_e^I a^I(p) + n_{m,I} a_{D,I}(p)\right) + \sum_A s_A m^A.$$

This is a consequence of a surface-charge operator relation, not merely of mutual commutativity. Two states with the same p and the same full charge γ can still have different energies, spins, particle content, and BPS properties; they nevertheless have the same central-charge eigenvalue.

Positivity of the supersymmetry anticommutator in the rest frame gives the BPS bound

$$M \geq |Z_\gamma|,$$

in a normalization in which the conventional proportionality factor is one. A BPS state saturates this bound and is annihilated by part of the supersymmetry algebra.

Why p and γ determine $Z_\gamma(p)$

$$\widehat{Z} \text{ is central} \Longrightarrow \widehat{Z} \text{ is scalar on an irreducible supermultiplet}$$

is only Schur's lemma. The stronger field-theory input is

$$\widehat{Z} = a^I(p)\widehat{n}_{e,I} + a_{D,I}(p)\widehat{n}_m^I + m^A\widehat{s}_A.$$

The vacuum fixes the asymptotic scalar coefficients; the charge sector fixes the flux eigenvalues. Only after this boundary-charge identity is used does (p, γ) determine the eigenvalue.

3.5 Electromagnetic charge lattice and special coordinates

At each $p \in C^\circ$, the allowed electric and magnetic charges form an integral lattice

$$(\Gamma_{\text{em}})_p \simeq \mathbb{Z}^{2r}$$

with the antisymmetric Dirac pairing

$$\langle\ ,\ \rangle : (\Gamma_{\text{em}})_p \times (\Gamma_{\text{em}})_p \longrightarrow \mathbb{Z}.$$

The lattices vary locally constantly over C° and therefore form an integral local system

$$\Gamma_{\text{em}} \longrightarrow C^\circ.$$

Parallel transport around a nontrivial loop can return a charge basis transformed by electric–magnetic monodromy.

On a simply connected patch, and assuming a principal polarization for simplicity, choose a symplectic basis

$$e_I,\ m^I, \qquad \langle e_I, m^J\rangle = \delta_I{}^J, \qquad \langle e_I, e_J\rangle = \langle m^I, m^J\rangle = 0.$$

More general global forms and line-operator spectra can modify the integral polarization while leaving the local discussion intact.

The central charge is a holomorphic section of the dual local system,

$$Z \in H^0\left(C^\circ, \Gamma_{\text{em}}^\vee \otimes_{\mathbb{Z}} \mathcal{O}_{C^\circ}\right),$$

meaning that $Z_\gamma(p)$ varies holomorphically with p for every locally constant charge γ. Define its components by

$$a^I := Z_{e_I}, \qquad a_{D,I} := Z_{m^I}.$$

Then, for

$$\gamma = \sum_I \left(n_e^I e_I + n_{m,I} m^I\right), \qquad n_e^I, n_{m,I} \in \mathbb{Z},$$

one has

$$Z_\gamma = \sum_I \left(n_e^I a^I + n_{m,I} a_{D,I}\right).$$

The integers $n_e^I, n_{m,I}$ are charges. The quantities $a^I, a_{D,I}$ are continuous complex central-charge values of basis charges and have mass dimension one; they are not integer charges. In a local electric

Lagrangian frame, a^I is also the VEV of the complex scalar in the I-th Abelian vector multiplet. One can choose the local electric polarization so that

$$da^1, \dots, da^r$$

are linearly independent; the a^I are then local holomorphic *special coordinates* on C°.

This integral-charge/local-period structure is part of the physical consistency conditions on Coulomb geometry; see [9] for a systematic discussion in rank one.

3.6 Rigid special Kähler geometry

Proposition 3.11 (Local special geometry)**.** *On C°, the two-derivative $\mathcal{N} = 2$ Abelian effective action makes the period map*

$$\Pi = \begin{pmatrix} a_{D,I} \\ a^I \end{pmatrix}$$

a holomorphic Lagrangian immersion into the local complex symplectic period space. Equivalently,

$$\sum_I da^I \wedge da_{D,I} = 0.$$

Consequently, on a sufficiently small electric coordinate patch there is a holomorphic prepotential $\mathcal{F}(a)$ such that

$$a_{D,I} = \frac{\partial \mathcal{F}}{\partial a^I},$$

and the effective gauge-coupling matrix is

$$\tau_{IJ} = \frac{\partial a_{D,I}}{\partial a^J} = \frac{\partial^2 \mathcal{F}}{\partial a^I \partial a^J}.$$

It is symmetric and satisfies $\operatorname{Im}\tau > 0$ *wherever the kinetic terms are positive and nondegenerate.*

An intrinsic mathematical formulation of rigid special Kähler geometry and its relation to algebraic integrable systems is given by Freed [7].

The Lagrangian condition makes the holomorphic one-form

$$\vartheta = a_{D,I}\, da^I$$

closed. The local Poincaré lemma then gives $\vartheta = d\mathcal{F}$, explaining why the prepotential exists locally. The Kähler potential and metric can be written

$$K = \operatorname{Im}\left(\overline{a^I} a_{D,I}\right), \qquad g_{I\bar{J}} = \partial_I \partial_{\bar{J}} K = \operatorname{Im}\tau_{IJ},$$

so

$$ds^2 = \operatorname{Im}\tau_{IJ}\, da^I d\bar{a}^J.$$

The same τ_{IJ} controls the Maxwell couplings, theta angles, and scalar kinetic terms.

This geometry is "special" because it is more than an ordinary Kähler manifold: it also carries the flat symplectic structure supplied by the electromagnetic local system. In a special frame,

$$x^I = \operatorname{Re} a^I, \qquad y_I = \operatorname{Re} a_{D,I}$$

are local flat symplectic affine coordinates. Equivalently, the geometry can be described by a flat torsion-free symplectic connection compatible with the complex and Kähler structures.

Neither the symplectic basis nor the prepotential is usually global. Around a physical singular locus,

$$\Pi \longmapsto M\Pi, \qquad M \in Sp(2r, \mathbb{Z})$$

in the principally polarized case, while the charge basis transforms contragrediently so that Z_γ is unchanged. The globally meaningful data are the charge local system, its pairing, the holomorphic central-charge section, and the induced metric.

The gauge-invariant chiral-ring coordinates

$$u_i = \langle O_i \rangle$$

should not be confused with a^I. The u_i are intrinsic algebraic functions on C, whereas $a^I = a^I(u)$ are generally multivalued periods of dimension one. For example, in the semiclassical region of rank-one $SU(2)$ theory,

$$u = \left\langle \operatorname{Tr} \phi^2 \right\rangle \sim a^2,$$

and the Weyl transformation $a \mapsto -a$ leaves u invariant.

3.7 Seiberg–Witten realization and degeneration

When an ordinary Seiberg–Witten curve description exists, it consists of a family of curves and a meromorphic differential, as in the original rank-one solutions [5,6],

$$\pi : \Sigma \longrightarrow C, \qquad \lambda_{\mathrm{SW}}.$$

For $u \in C^\circ$, the fiber Σ_u is smooth. A locally chosen symplectic cycle basis $A_I, B^I \in H_1(\Sigma_u, \mathbb{Z})$ realizes the charge basis, and

$$a^I(u) = \oint_{A_I} \lambda_{\mathrm{SW}}, \qquad a_{D,I}(u) = \oint_{B^I} \lambda_{\mathrm{SW}}, \qquad Z_\gamma(u) = \oint_\gamma \lambda_{\mathrm{SW}}.$$

For more general higher-rank theories, the appropriate object can be an algebraic integrable system rather than a single preferred curve; its periods encode the same special Kähler data.

At the physical discriminant, a cycle can degenerate and its period can tend to zero:

$$Z_\gamma(u) \longrightarrow 0.$$

When a BPS state of charge γ is present, its mass tends to zero and the Abelian effective action that had integrated it out ceases to be complete. The Seiberg–Witten family can often be extended over this locus with singular fibers, but the nonsingular rigid special Kähler effective geometry is defined only on C°. The conformal origin of an interacting SCFT is typically among these special degenerations.

Example 3.12 (Lagrangian gauge theory)**.** In a Lagrangian $\mathcal{N} = 2$ gauge theory, the vector multiplet contains a complex adjoint scalar ϕ. A generic Coulomb vacuum has

$$\left\langle \phi \right\rangle \neq 0,$$

which can be placed in a Cartan subalgebra and breaks a rank-r gauge group G to its maximal torus T, locally

$$T \simeq U(1)^r.$$

For a root α, the corresponding non-Cartan W-boson has mass proportional to

$$|\alpha(\langle\phi\rangle)|.$$

This is nonzero at a generic point, so only the r Cartan photons remain massless. Gauge-invariant polynomials such as

$$u_k = \langle \operatorname{Tr} \phi^k \rangle$$

provide Coulomb-branch coordinates.

The intrinsic definition above keeps the gauge-invariant operators $\operatorname{Tr} \phi^k$ but does not require the existence of the elementary field ϕ.

Remark 3.13. The Coulomb chiral ring is often freely generated, but freeness is not part of the definition. If the generators obey relations, then C is a nontrivial affine variety rather than an affine space.

4 The Higgs Branch

The Higgs-side dictionary

A highest-weight $\widehat{\mathcal{B}}_R$ primary gives a holomorphic Higgs-branch coordinate after one complex structure is chosen. The full real branch is a hyperkähler cone: $SU(2)_R$ rotates its complex structures, while flavor symmetry preserves each one and is generated by a triplet of moment maps.

4.1 Higgs-branch operators

Definition 4.1 ($\widehat{\mathcal{B}}_R$ primary and Higgs-branch chiral operator)**.** Let $R \in \frac{1}{2}\mathbb{Z}_{\geq 0}$. A $\widehat{\mathcal{B}}_R$ superconformal primary is a Lorentz scalar with

$$j_1 = j_2 = 0, \qquad r = 0, \qquad \Delta = 2R,$$

transforming in the spin-R representation of $SU(2)_R$. In doublet-index notation it can be written

$$\mathcal{O}^{(i_1 \cdots i_{2R})}, \qquad i_a = 1, 2,$$

with all $2R$ indices symmetrized. In the convention where the index 1 has R_3 weight $+\frac{1}{2}$, its highest-weight component is

$$\mathcal{O}_{\mathrm{HW}} := \mathcal{O}^{11\cdots 1}, \qquad R_3 \mathcal{O}_{\mathrm{HW}} = R\mathcal{O}_{\mathrm{HW}}.$$

It obeys the shortening conditions

$$Q^1_\alpha \mathcal{O}_{\mathrm{HW}}(0) = 0, \qquad \widetilde{Q}_{2\dot{\alpha}} \mathcal{O}_{\mathrm{HW}}(0) = 0, \qquad \alpha, \dot{\alpha} = 1, 2.$$

This highest-weight component is called a *Higgs-branch chiral operator*.

The $\widehat{\mathcal{B}}_R$ notation and null-state structure follow the superconformal multiplet classification [1]; the same protected multiplets play a central role in the four-dimensional chiral-algebra construction [2].

The $2R$ indices are a notation for the standard realization

$$\mathrm{Sym}^{2R}(\mathbf{2})$$

of the spin-R irreducible representation of $SU(2)$. A completely symmetric tensor with $2R$ binary indices has $2R + 1$ independent components, with R_3 weights

$$R, R-1, \ldots, -R.$$

Thus $2R$ is the number of fundamental doublet indices, not the number of additional physical degrees of freedom. For example,

$$R = \tfrac{1}{2} : O^i, \qquad R = 1 : O^{(ij)}, \qquad R = \tfrac{3}{2} : O^{(ijk)}.$$

The supercharge annihilation is an independent shortening condition, not a consequence of being highest weight alone. In radial quantization, let

$$|O\rangle = O_{\mathrm{HW}}(0)|0\rangle.$$

Because $(Q^I_\alpha)^\dagger = S^\alpha_I$ and $(\widetilde{Q}_{I\dot{\alpha}})^\dagger = \widetilde{S}^{I\dot{\alpha}}$, the superconformal algebra gives, up to positive convention-dependent factors,

$$\left\| Q^1_\alpha |O\rangle \right\|^2 \propto (\Delta - 2R - r)\|O\|^2,$$

and

$$\left\| \widetilde{Q}_{2\dot{\alpha}} |O\rangle \right\|^2 \propto (\Delta - 2R + r)\|O\|^2.$$

For $\Delta = 2R$ and $r = 0$, these descendants have zero norm and are set to zero in a unitary theory. By contrast, for $R > 0$ the complementary descendants $Q^2_\alpha|O\rangle$ and $\widetilde{Q}_{1\dot{\alpha}}|O\rangle$ have nonzero norm in general. Hence the highest-weight operator is annihilated by four of the eight Poincaré supercharges and is 1/2-BPS. Raising or lowering $SU(2)_R$ indices may exchange the displayed labels 1 and 2 in other conventions.

What actually proves the shortening

$$\Delta = 2R, \qquad r = 0 \Longrightarrow \|Q^1 O_{\mathrm{HW}}\|^2 = \|\widetilde{Q}_2 O_{\mathrm{HW}}\|^2 = 0 \Longrightarrow Q^1 O_{\mathrm{HW}} = \widetilde{Q}_2 O_{\mathrm{HW}} = 0.$$

The first implication uses the superconformal anticommutators; the second uses unitarity and quotienting by null states. Highest weight by itself is not enough.

The cases $\widehat{\mathcal{B}}_0$, $\widehat{\mathcal{B}}_{1/2}$, and $\widehat{\mathcal{B}}_1$ contain, respectively, the identity, a free-hypermultiplet scalar, and a flavor moment-map primary. Selecting the highest weight chooses an R_3 direction and hence one complex structure from the hyperkähler S^2 of complex structures; in that complex structure O_{HW} is holomorphic.

Definition 4.2 (Higgs-branch chiral ring)**.** The *Higgs-branch chiral ring* $\mathcal{R}_{\mathrm{HB}}$ is the commutative ring formed by highest-weight Higgs-branch operators, modulo their operator relations.

Definition 4.3 (Higgs branch)**.** The *Higgs branch* is the family of $\mathcal{N} = 2$ Poincaré-supersymmetric vacua parameterized by VEVs of Higgs-branch chiral operators, with Coulomb-type order parameters set to zero. In a chosen complex structure,

$$\mathcal{H} = \mathrm{Spec}\,\mathcal{R}_{\mathrm{HB}}.$$

As for the Coulomb branch, an ordinary complex-valued Higgs vacuum corresponds to a character $\mathcal{R}_{\mathrm{HB}} \to \mathbb{C}$ and hence to a maximal ideal. The full $\mathrm{Spec}\,\mathcal{R}_{\mathrm{HB}}$ additionally retains its scheme and local-ring structure.

4.2 Hyperkähler geometry

Definition 4.4 (Hyperkähler manifold)**.** A *hyperkähler manifold* is a Riemannian manifold (M, g) equipped with three integrable complex structures I, J, K satisfying

$$I^2 = J^2 = K^2 = IJK = -1,$$

such that g is Kähler with respect to each complex structure.

The associated Kähler forms are

$$\omega_I(X, Y) = g(IX, Y), \qquad \omega_J(X, Y) = g(JX, Y), \qquad \omega_K(X, Y) = g(KX, Y).$$

There are not merely three isolated complex structures. For every unit vector $\vec{n} = (n_1, n_2, n_3) \in S^2$,

$$I_{\vec{n}} = n_1 I + n_2 J + n_3 K$$

also satisfies $I_{\vec{n}}^2 = -1$. Thus a hyperkähler manifold carries an $S^2 \simeq \mathbb{CP}^1$ family of complex structures.

Proposition 4.5. *The smooth locus of the Higgs branch is hyperkähler. Its real dimension is divisible by four:*

$$\dim_{\mathbb{R}} \mathcal{H} = 4d_H,$$

where d_H is called the quaternionic dimension:

$$d_H = \dim_{\mathbb{H}} \mathcal{H}.$$

The relation between extended supersymmetric sigma models, hyperkähler geometry, and hyperkähler quotients was developed systematically by Hitchin, Karlhede, Lindström, and Roček [8].

Definition 4.6 (Hyperkähler cone)**.** A *hyperkähler cone* is a hyperkähler space whose smooth locus has a closed homothetic vector field χ satisfying

$$\nabla_X \chi = X, \qquad \text{and hence} \qquad \mathcal{L}_\chi g = 2g.$$

Locally its metric has cone form

$$g = dr^2 + r^2 g_Y, \qquad \chi = r\frac{\partial}{\partial r}.$$

The vector fields

$$k_I = I\chi, \qquad k_J = J\chi, \qquad k_K = K\chi$$

are Killing and generate an $SU(2)$ action that rotates the three complex structures. The Higgs branch of a four-dimensional $\mathcal{N} = 2$ SCFT is such a cone, possibly singular at its tip and along lower-dimensional strata.

The radial homothety is the geometric action of dilatations on Higgs-branch VEVs. The rotating $SU(2)$ is the geometric realization of $SU(2)_R$: on the triple (I, J, K) its action factors through $SO(3) = SU(2)/\{\pm 1\}$. After choosing, say, I as the holomorphic complex structure, the subgroup $U(1) \subset SU(2)_R$ preserving I combines with positive real scaling to give the holomorphic $\mathbb{C}^*$ cone action.

Flavor symmetry behaves differently. It belongs to the full SCFT, not only to its Higgs branch, but its action on $\mathcal{H}$ is by tri-holomorphic isometries. If k_a is the vector field generated by a flavor generator, then

$$\mathcal{L}_{k_a} g = 0, \qquad \mathcal{L}_{k_a} I = \mathcal{L}_{k_a} J = \mathcal{L}_{k_a} K = 0.$$

Thus $SU(2)_R$ rotates the complex structures, whereas flavor symmetry preserves each one separately.

4.3 The flavor-current multiplet and moment maps

Let F be a continuous flavor group. For each generator $T^a \in \mathfrak{f}$ there is a conserved current

$$j^a_\mu, \qquad \partial^\mu j^a_\mu = 0, \qquad F^a = \int d^3x\, j^a_0.$$

In a four-dimensional CFT, a conserved vector current has

$$\Delta(j^a_\mu) = 3, \qquad (j_1, j_2) = \left(\tfrac{1}{2}, \tfrac{1}{2}\right).$$

Because a flavor charge commutes with the full superconformal algebra, this current is an $SU(2)_R \times U(1)_r$ singlet.

The current is not the superconformal primary of its multiplet. Conservation, shortening, and the condition $[F^a, Q] = 0$ place an ordinary flavor current in a unique $\mathcal{N} = 2$ current multiplet [1], conventionally denoted

$$\widehat{\mathcal{B}}_1.$$

Its superconformal primary is a Lorentz scalar $\mu^{a(ij)} = \mu^{a(ji)}$ with

$$\Delta = 2, \qquad R = 1, \qquad r = 0.$$

The current occurs at the $Q\widetilde{Q}$ descendant level, schematically

$$j^a_{\alpha\dot\alpha} \sim Q^i_\alpha \widetilde{Q}^j_{\dot\alpha} \mu^a_{ij} + \text{terms selecting the conserved conformal primary}.$$

The dimension count $2 + \frac{1}{2} + \frac{1}{2} = 3$ and the Lorentz product $(\frac{1}{2}, 0) \otimes (0, \frac{1}{2})$ explain how the vector can occur, but they do not by themselves prove uniqueness. The uniqueness statement uses the classification of shortened superconformal multiplets containing conserved currents. In another common notation the same multiplet is

$$B_1\overline{B}_1[0;0]^{(2;0)}_2,$$

where the $SU(2)_R$ Dynkin label 2 means spin $R = 1$.

By contrast, the stress-tensor multiplet contains the $SU(2)_R \times U(1)_r$ currents. Their charges rotate the supercharges and therefore are not flavor charges.

The moment-map primary and its protected correlators are also reviewed from the chiral-algebra viewpoint in [2].

Definition 4.7 (Flavor moment-map operator)**.** The primary

$$\mu^{a(ij)} \in \mathrm{adj}(F) \otimes \mathbf{3}_{SU(2)_R}$$

of the $\widehat{\mathcal{B}}_1$ flavor-current multiplet is called the *flavor moment-map operator*. Its highest-weight component, for example $\mu^{a(11)}$, is a Higgs-branch chiral operator.

The terminology has a direct geometric meaning. Let k_a generate the flavor action on the smooth Higgs branch. Since this action preserves all three Kähler forms, it has a triplet of moment maps satisfying, up to an overall sign convention,

$$d\mu^a_I = \iota_{k_a}\omega_I, \qquad d\mu^a_J = \iota_{k_a}\omega_J, \qquad d\mu^a_K = \iota_{k_a}\omega_K.$$

These three real functions are the geometric realization of the three components of $\mu^{a(ij)}$. In complex structure I, one combines

$$\Omega_I = \omega_J + i\omega_K, \qquad \mu^a_{\mathbb{C}} = \mu^a_J + i\mu^a_K,$$

so that

$$d\mu^a_{\mathbb{C}} = \iota_{k_a}\Omega_I.$$

The holomorphic function $\mu^a_{\mathbb{C}}$ is the chosen highest-weight component in the Higgs chiral ring.

Higgs-branch operators may transform in nontrivial flavor representations, so flavor transformations can move points of $\mathcal{H}$ and a Higgs VEV may break F to the stabilizer of the chosen point. This is why flavor moment maps naturally appear in the Higgs-branch discussion. Flavor symmetry still belongs to the whole SCFT: it acts on BPS states above the Coulomb branch and its background masses deform the Coulomb special geometry, even though continuous flavor acts trivially on the pure Coulomb base itself.

Example 4.8 (Lagrangian hyperkähler quotient)**.** In a Lagrangian gauge theory with hypermultiplet scalar space $\mathcal{X}$ and gauge group G, the classical Higgs branch is

$$\mathcal{H} = \mu_G^{-1}(0)/G,$$

where μ_G is the triplet of *gauge* moment maps. These are constraints used to remove gauge redundancy: one imposes $\mu_G = 0$ and divides by G. Equivalently, in a chosen complex structure one imposes the complex gauge-moment-map equation and takes the quotient by $G_{\mathbb{C}}$, with the appropriate stability condition.

This must be distinguished from a flavor moment map μ_F. A surviving flavor group is a genuine global symmetry of the quotient, and μ_F descends to a gauge-invariant physical function on $\mathcal{H}$. Thus gauge moment maps construct the Higgs branch, while flavor moment maps are operators living on it. The intrinsic SCFT definition replaces elementary hypermultiplet fields by these gauge-invariant operators and their ring relations.

5 Coulomb, Higgs, and Mixed Branches

	Coulomb branch	Higgs branch	Mixed branch
Order parameters	Coulomb chiral operators	$\widehat{\mathcal{B}}_R$ operators	Both types
$SU(2)_R$ quantum number	$R = 0$	generally $R > 0$	both sectors present
$U(1)_r$ quantum number	nonzero, convention-dependent	$r = 0$	both sectors present
Natural geometry	rigid special Kähler on C°	hyperkähler cone	stratified combination of Coulomb and Higgs directions
Generic massless fields	r Abelian vector multiplets; neutral hypers may coexist	hypermultiplet moduli, possibly with a residual sector	vector and hypermultiplet moduli

Definition 5.1 (Mixed branch)**.** A *mixed branch* is a family of supersymmetric vacua in which compatible Coulomb-branch and Higgs-branch operators simultaneously acquire nonzero VEVs.

Definition 5.2 (Enhanced Coulomb branch)**.** An *enhanced Coulomb branch* is a mixed branch whose generic point lies over the Coulomb base and contains additional neutral massless hypermultiplet directions. Locally over a regular patch it has the form

$$C^{\circ} \times \mathbb{H}^{h}$$

or, more generally, a hyperkähler fibration over C°. The integer h counts neutral hypermultiplets and does not change the Coulomb rank r.

Enhanced Coulomb branches and their role in rank-one SCFT geometry are analyzed in [10].

A mixed branch is usually stratified. Near a sufficiently generic smooth point of a stratum, its massless directions often locally separate into Coulomb-type and Higgs-type factors, although global product decompositions need not exist.

Remark 5.3. Statements such as "the Coulomb branch is given by $q = 0$" and "the Higgs branch is given by $\phi = 0$" belong to a particular Lagrangian presentation. The intrinsic distinction is instead determined by the superconformal multiplets of the operators acquiring VEVs.

6 Expanding Around a Vacuum

The infrared question

Fixing a vacuum chooses a superselection sector and generates mass thresholds. One first constructs a finite-energy EFT and only then takes its strict infrared limit. The tangent multiplets become free at a smooth point, but the complete answer may include an additional residual factor.

6.1 What an expansion around a vacuum means

Choose a vacuum $p \in \mathcal{M}_{\mathrm{vac}}$ and a local operator with VEV

$$\langle \mathcal{O} \rangle_p = v.$$

One writes schematically

$$\mathcal{O}(x) = v + \delta\mathcal{O}(x),$$

where $\delta\mathcal{O}$ represents fluctuations around the chosen vacuum.

Definition 6.1 (Low-energy effective field theory)**.** A *low-energy effective field theory* is the theory governing degrees of freedom whose characteristic energies satisfy

$$E \ll M,$$

where M is the mass scale of the degrees of freedom that have been removed from the description.

Definition 6.2 (Integrating out)**.** To *integrate out* a massive degree of freedom means to eliminate it from the explicit low-energy description while retaining its effects through effective interactions among the remaining light fields.

For a VEV v of dimension Δ, the natural separation scale is

$$M_v \sim |v|^{1/\Delta}.$$

At energies $E \ll M_v$, massive states created by the symmetry breaking may be integrated out.

Definition 6.3 (Massless degree of freedom)**.** A *massless degree of freedom* is an excitation whose energy can become arbitrarily small as its spatial momentum tends to zero. Such fields remain in the deep infrared effective theory.

6.2 The conformal vacuum

At the tip of every branch,

$$\langle O_i \rangle = 0$$

for all positive-dimension order parameters. Expanding around this vacuum gives the original SCFT itself. This is the unique point at which the full superconformal symmetry is manifest in the vacuum.

Proposition 6.4. *The original interacting SCFT is recovered at the conformal origin, not at a generic nonzero-VEV vacuum.*

6.3 Finite-energy EFT versus the strict IR fixed point

A VEV is not a coupling deformation

Vacuum selection keeps the microscopic theory $\mathcal{T}$ fixed and chooses a state $|\Omega_p\rangle$ in which observables are evaluated. A coupling deformation instead changes the action, schematically

$$S \longmapsto S + \lambda \int d^4x\, O(x).$$

Thus a VEV means "same theory, different state", whereas changing λ moves in theory space. These operations can lead to similar-looking scales but are conceptually different.

In infinite volume, different vacua are superselection sectors selected by boundary conditions at spatial infinity. Wilsonian coarse graining is then performed inside the chosen sector, so the infrared endpoint naturally depends on the pair

$$(\mathcal{T}, p) \longmapsto \mathcal{T}_{\mathrm{IR},p}.$$

Different vacua of the same microscopic theory can consequently have different infrared fixed points without the VEV becoming a coupling.

A nonzero VEV nevertheless generates a scale

$$M_p \sim |v|^{1/\Delta},$$

or, on a Coulomb branch, the collection of charged thresholds $|Z_\gamma(p)|$. At energies $E \ll M_p$, massive states can be integrated out and one obtains a Wilsonian effective action

$$S_{\mathrm{EFT},p} = S_2 + \sum_{d>4} \frac{c_d}{M_p^{d-4}} \int d^4x\, \mathcal{O}_d.$$

At finite nonzero E/M_p, this action still remembers the VEV scale and is generally not conformal.

The *strict IR fixed point* is obtained by taking

$$E/M_p \longrightarrow 0.$$

Irrelevant interactions disappear in this limit. The resulting fixed point can be a free SCFT, an interacting SCFT, or a product of free and interacting factors. It need not equal the original SCFT at the conformal vacuum.

There is no contradiction between spontaneous breaking of the original dilatation symmetry and emergent conformal symmetry of the fluctuation theory. If

$$a(x) = a_0 + \delta a(x),$$

the original dilatation acts on δa with an inhomogeneous term proportional to a_0, so the chosen vacuum is not invariant. The Gaussian strict-IR action can instead possess an emergent dilatation acting homogeneously on δa.

Finally, a massless degree of freedom need not be a modulus. The tangent space to a branch counts scalar fluctuations whose constant values move the vacuum, but an additional gapless or interacting sector can have no such vacuum direction. Consequently, proving that the tangent-mode sector is free does not prove that the complete IR theory contains nothing else.

The order of operations

$$\text{choose } p \Longrightarrow \text{identify the mass scale } M_p \Longrightarrow \text{integrate out states with } M \gtrsim M_p$$
$$\Downarrow$$
$$S_{\mathrm{EFT},p} \xrightarrow{E/M_p \to 0} \mathcal{T}_{\mathrm{IR},p}.$$

The finite-energy EFT generally remembers M_p and is not conformal. The strict IR fixed point is the limiting theory after irrelevant interactions have disappeared.

6.4 A generic Coulomb-branch vacuum

Let $p \in \mathcal{C}^\circ$ be a generic physical regular point of a rank-r Coulomb branch. The low-energy massless fields contain r Abelian $\mathcal{N}=2$ vector multiplets. In a chosen electric frame the gauge group is locally written

$$U(1)^r.$$

More invariantly, the local gauge algebra is $\mathfrak{u}(1)^r$; discrete quotients and the spectrum of genuine line operators are additional global data.

The special Kähler action describes the vector-moduli sector, not automatically the complete massless theory. Denote this part by S_{vec}. In local special coordinates, its bosonic terms include schematically

$$S_{\mathrm{vec}} \supset \int d^4x\ \mathrm{Im}\,\tau_{IJ}(a) \left(F^I_{\mu\nu}F^{J\,\mu\nu} + \partial_\mu a^I \partial^\mu \overline{a}^J\right) + \cdots .$$

Expanding around $a^I = a_0^I$ gives

$$\tau_{IJ}(a) = \tau_{IJ}(a_0) + \partial_K \tau_{IJ}(a_0)\,\delta a^K + \cdots .$$

The quadratic term describes free Maxwell multiplets with constant couplings. The first interaction contains terms such as

$$\partial_K \tau_{IJ}(a_0)\,\delta a^K F^I_{\mu\nu} F^{J\,\mu\nu}.$$

Since $[\delta a] = 1$ and $[F] = 2$, this is a dimension-five interaction whose coefficient has dimension -1 and is suppressed by the VEV scale. Higher powers and higher-derivative terms are likewise irrelevant. Therefore

$$(S_{\text{vec}})_{\text{strict IR}} = \{r \text{ free } \mathcal{N}=2 \text{ Maxwell multiplets}\}.$$

This conclusion concerns only the vector-moduli sector. The complete low-energy action may have the schematic form

$$S_{\text{IR},p} = S_{\text{vec}} + S_{\text{neutral}} + S_{\text{mix}}.$$

If the mixing interactions are irrelevant, the strict IR fixed point factorizes as

$$\mathcal{T}_{\text{IR},p} \simeq \{r \text{ free Abelian vector multiplets}\} \times \mathcal{T}_{\text{neutral},p}.$$

Physical regularity excludes additional massless states carrying the electromagnetic charges of the $U(1)^r$ sector, but it does not exclude neutral massless fields. Moreover, an additional massless vector modulus would increase $\dim_{\mathbb{C}} C$ and is therefore already counted in r. The residual neutral factor can contain free neutral hypermultiplets, as on an enhanced Coulomb branch, or a decoupled rank-zero gapless sector. A genuinely interacting residual factor at a generic point is possible in principle but is more special than the usual free-vector situation.

Example 6.5 ($\mathcal{N}=4$ Yang–Mills viewed as an $\mathcal{N}=2$ theory)**.** An $\mathcal{N}=4$ vector multiplet decomposes into an $\mathcal{N}=2$ vector multiplet and an adjoint hypermultiplet. At a generic rank-r Coulomb VEV, the non-Cartan fields are massive, while the Cartan fields give

$$r \text{ free } \mathcal{N}=2 \text{ vectors} \quad + \quad r \text{ free neutral hypermultiplets}.$$

Equivalently, the strict IR contains r free Abelian $\mathcal{N}=4$ multiplets. This is a standard example in which the Coulomb vector sector is not the whole massless theory.

6.5 A generic Higgs-branch vacuum

Let p be a smooth point of the Higgs branch. The scalar fluctuations tangent to $\mathcal{H}$ form hypermultiplets. Their two-derivative action is a nonlinear sigma model; to emphasize that this is only the moduli sector, write

$$S_{\text{moduli}} = \int d^4x\, g_{A\overline{B}}(X)\partial_\mu X^A \partial^\mu \overline{X}^{\overline{B}} + \cdots .$$

Choose normal coordinates near p and write

$$X^A = X_0^A + \delta X^A.$$

Then

$$g_{A\overline{B}}(X) = g_{A\overline{B}}(X_0) + O(\delta X^2).$$

The leading term describes free hypermultiplet fluctuations associated with $T_p\mathcal{H}$. The first curvature correction is schematically

$$R_{A\overline{B}C\overline{D}}(X_0)\,\delta X^C \delta\overline{X}^{\overline{D}} \partial_\mu \delta X^A \partial^\mu \delta\overline{X}^{\overline{B}}.$$

It has dimension six in four spacetime dimensions and is irrelevant. Hence the tangent hypermultiplets become free in the strict IR.

Again, this proves a statement about the moduli sector rather than the complete IR theory. In general,

$$S_{\mathrm{IR},p} = S_{\mathrm{moduli}} + S_{\mathrm{residual}} + S_{\mathrm{mix}},$$

and, when the mixing flows to zero,

$$\mathcal{T}_{\mathrm{IR},p} \simeq \{\text{free hypermultiplets from } T_p\mathcal{H}\} \times \mathcal{T}_{\mathrm{residual},p}.$$

Smoothness of $\mathcal{H}$ only controls the vacuum directions parameterized by the Higgs chiral ring. It cannot exclude a massless sector with no Higgs modulus.

For example, if

$$\mathcal{T} = \mathcal{T}_1 \times \mathcal{T}_2,$$

where only $\mathcal{T}_1$ has a Higgs branch, then a VEV in $\mathcal{T}_1$ leaves $\mathcal{T}_2$ untouched. The infrared theory can therefore contain free tangent hypers times the residual theory $\mathcal{T}_2$, even at a smooth point of $\mathcal{H}$.

Residual vector multiplets require more care. At a generic point of a pure Higgs branch of an indecomposable theory, one normally expects no unbroken dynamical vector sector. If a gauge group is only partially Higgsed and massless vectors remain, their scalars generally supply Coulomb directions; this indicates a mixed branch or a boundary between Higgs and mixed strata. A decoupled vector factor is another possibility. Thus residual interacting sectors are compatible with a smooth Higgs branch, but residual vectors should not be asserted generically without this qualification.

If no residual sector is present, the complete strict-IR theory at a generic smooth Higgs point consists only of free hypermultiplets.

6.6 Singular vacua

Definition 6.6 (Geometric and physical singularity)**.** A point $p \in \mathcal{M}_{\mathrm{vac}}$ is *geometrically singular* if its local algebraic or differential moduli-space geometry fails to be smooth. A Coulomb vacuum is *physically singular* if it lies in Δ_{phys}, so the Abelian vector-multiplet effective description is singular or incomplete because additional charged degrees of freedom become massless.

The two notions are related but not identical. A geometric singularity often signals extra massless degrees of freedom, but a physically singular Coulomb point can lie on an algebraically smooth base. At the Coulomb discriminant, a charge γ can obey

$$Z_\gamma(p) = 0.$$

If a BPS state of this charge is present, it becomes massless and cannot be integrated out. A single mutually local collection can admit another weakly coupled description, while simultaneous mutually nonlocal massless states can produce an interacting fixed point of Argyres–Douglas type [11]. At a singular Higgs-branch locus, additional interacting directions or unbroken sectors may likewise appear.

The infrared theory at such a point can be

$$\mathcal{T}_{\mathrm{IR},p} = \mathcal{T}'_{\mathrm{SCFT}} \times \mathcal{T}_{\mathrm{free}},$$

where $\mathcal{T}'_{\mathrm{SCFT}}$ may be a new interacting SCFT and $\mathcal{T}_{\mathrm{free}}$ contains decoupled free vector or hypermultiplets.

Remark 6.7. An interacting SCFT appearing at a singular point need not be identical to the original SCFT at the common origin. It may have lower rank, different flavor symmetry, different Coulomb dimensions, or different anomaly coefficients. The microscopic theory is unchanged by selecting p, but the strict IR fixed point of the sector $(\mathcal{T}, p)$ can be different because massive thresholds and the set of remaining massless states depend on the vacuum.

6.7 Summary by vacuum type

Vacuum	Preserved symmetry	Theory of low-energy fluctuations
Conformal origin	full $\mathcal{N} = 2$ superconformal symmetry	the original interacting SCFT
Generic physical regular Coulomb vacuum	$\mathcal{N} = 2$ Poincaré supersymmetry	r free Abelian vector multiplets, possibly times a neutral residual factor
Generic smooth Higgs vacuum	$\mathcal{N} = 2$ Poincaré supersymmetry	free tangent hypermultiplets, possibly times a residual factor; vectors indicate mixed or decoupled sectors
Generic smooth mixed vacuum	$\mathcal{N} = 2$ Poincaré supersymmetry	vector and hypermultiplet moduli, possibly times a residual sector
Singular vacuum	at least $\mathcal{N} = 2$ Poincaré supersymmetry	extra massless states; possibly a new interacting SCFT times free fields

7 Conceptual Dictionary

SCFT
A quantum field theory invariant under both conformal transformations and supersymmetry.

Local operator
An operator inserted at a spacetime point. Its products and correlation functions encode local measurements.

Operator spectrum
The collection of local operators organized by scaling dimension, Lorentz spin, R-symmetry, flavor symmetry, and superconformal multiplet.

OPE
The short-distance expansion of a product of local operators into a sum of local operators.

Cluster decomposition
: Factorization of correlators at infinite spacelike separation in a pure vacuum. Together with protected position independence, it makes the VEV map multiplicative.

VEV
: The expectation value of an operator in a chosen vacuum.

Vacuum
: A translationally invariant state of minimum energy. A supersymmetric vacuum is additionally annihilated by the Poincaré supercharges.

Modulus
: A continuous parameter labeling inequivalent vacua.

Massless degree of freedom versus modulus
: A modulus becomes a massless scalar fluctuation, but the converse need not hold. A gapless sector can have an isolated vacuum and therefore contribute no tangent direction to the vacuum moduli space.

Vacuum moduli space
: The space formed by all continuously connected supersymmetric vacua.

Branch
: A distinguished component or stratum of the vacuum moduli space, classified by which protected operators acquire VEVs.

Chiral ring
: A protected commutative ring of specified scalar primaries, with multiplication defined by the protected OPE. Its algebraic relations become equations for a moduli-space branch.

$\widehat{\mathcal{B}}_R$ multiplet
: A half-BPS multiplet whose scalar superconformal primary has $\Delta = 2R$, $r = 0$, and $SU(2)_R$ spin R. It may be written as a tensor with $2R$ symmetric doublet indices; its highest-weight component is a Higgs-branch chiral operator.

Flavor-current multiplet
: The $\widehat{\mathcal{B}}_1$ multiplet associated with a continuous flavor generator. Its scalar primary $\mu^{a(ij)}$ is an $SU(2)_R$ triplet, while the conserved current j^a_μ is a superconformal descendant.

Flavor moment map
: The $\widehat{\mathcal{B}}_1$ primary viewed as a triplet of Hamiltonians for the tri-holomorphic flavor action on the Higgs branch. It is a physical operator, unlike a gauge moment map used as a hyperkähler-quotient constraint.

Character and closed point
: A unital $\mathbb{C}$-algebra map $\chi_p : R \to \mathbb{C}$. Its kernel is the maximal ideal corresponding to the ordinary complex vacuum p.

Affine spectrum
: $\operatorname{Spec} R$ is the scheme of all prime ideals together with its local rings and structure sheaf. Its non-closed generic points are not additional ordinary complex vacua.

Coulomb branch
: The branch parameterized by VEVs of $SU(2)_R$-singlet Coulomb chiral primaries. Its physical regular locus is rigid special Kähler.

Higgs branch
: The branch parameterized by VEVs of highest-weight $\widehat{\mathcal{B}}_R$ operators. It is a hyperkähler cone: dilatations generate the homothety, $SU(2)_R$ rotates its sphere of complex structures, and flavor symmetry preserves each complex structure.

Mixed branch
A branch carrying compatible VEVs of both Coulomb and Higgs types.

Enhanced Coulomb branch
A mixed branch with neutral hypermultiplet directions fibered over a generic Coulomb base.

Electromagnetic charge lattice
The integral local system $\Gamma_{\text{em}} \to C^\circ$ of electric and magnetic charges, equipped with the Dirac pairing and electric–magnetic monodromy.

Supersymmetry central charge
The center operator $\widehat{Z}$ in the $4d$ $\mathcal{N} = 2$ Poincaré superalgebra. In a fixed vacuum it is a surface charge whose eigenvalue is determined by the full electric, magnetic, and flavor charge sector.

Special coordinates
The continuous complex period functions $a^I = Z_{e_I}$ and $a_{D,I} = Z_{m^I}$ associated with a local symplectic charge basis. They are not integer charges, and only the a^I are chosen as local coordinates in an electric frame.

Rigid special Kähler geometry
The Kähler metric together with the flat integral symplectic structure and holomorphic Lagrangian period map supplied by (Γ_{em}, Z). Locally it is encoded by a prepotential $\mathcal{F}$.

Seiberg–Witten geometry
A curve or algebraic-integrable-system realization of the charge local system and central-charge periods. Its smooth fibers describe C° and its degenerations encode the physical discriminant.

Effective field theory
A description valid below a chosen energy scale, containing only the degrees of freedom that remain light at that scale.

Infrared limit
The limit of physical processes at energies much smaller than all massive scales.

Strict IR fixed point
The limiting scale-invariant theory obtained after all irrelevant interactions of a finite-energy EFT have flowed away. It can be different from the original SCFT even when the scale arose by choosing a vacuum rather than by deforming the action.

Residual sector
Massless degrees of freedom not parameterized by the tangent directions of the branch under discussion. At a physical regular Coulomb point such a sector must be electromagnetically neutral; on a pure Higgs branch residual vectors usually signal a mixed or decoupled factor.

Free field theory
A theory whose leading action is quadratic and whose asymptotic excitations do not interact.

Interacting fixed point
A scale-invariant infrared or ultraviolet limit with nontrivial OPE coefficients and genuine interactions.

Singular vacuum
A geometrically singular point or a physically singular point where an effective description degenerates. The notions often coincide but need not do so.

8 Master Reasoning Map

The intrinsic construction can be summarized as

$$\begin{gathered}\text{SCFT operator spectrum and OPE}\\ \Downarrow\\ \text{specified protected scalar primaries and their protected OPE rings}\\ \Downarrow\\ \text{Ward-identity position independence plus cluster decomposition}\\ \Downarrow\\ \text{VEV characters } \chi_p : R \to \mathbb{C} \quad \Longrightarrow \quad \text{closed points of } \operatorname{Spec} R.\end{gathered}$$

The algebraic and low-energy constructions are distinct layers:

$$\boxed{\begin{array}{rl} \mathcal{R}_{\mathrm{CB}} \Longrightarrow \mathcal{C} = \operatorname{Spec} \mathcal{R}_{\mathrm{CB}} & \text{complex algebraic or scheme data,}\\ p \in \mathcal{C}^\circ \text{ and low-energy } \mathcal{N} = 2 \Longrightarrow \mathfrak{u}(1)^r,\ \Gamma_{\mathrm{em}},\ Z & \text{dynamical electromagnetic data,}\\ (\Gamma_{\mathrm{em}}, Z) \Longrightarrow (a^I, a_{D,I}),\ \mathcal{F},\ \tau,\ g & \text{rigid special Kähler data.}\end{array}}$$

In particular, the photons and Seiberg–Witten periods do not follow from $\operatorname{Spec} \mathcal{R}_{\mathrm{CB}}$ alone.

After choosing a vacuum p, one studies fluctuations by

$$\boxed{\text{SCFT} \xrightarrow{\text{choose a BPS VEV}} \mathcal{N} = 2 \text{ Poincaré vacuum} \xrightarrow{E \ll M_v} \text{infrared effective theory near } p.}$$

Thus the origin gives the original SCFT; a regular point gives free moduli fields, possibly with a residual sector; and a singular point can add massless states or a new interacting SCFT.

More precisely, branch geometry determines the tangent multiplets but does not by itself exhaust the massless spectrum:

$$\boxed{\mathcal{T}_{\mathrm{IR},p} \simeq \mathcal{T}_{\mathrm{tangent,free},p} \times \mathcal{T}_{\mathrm{residual},p}.}$$

For a physical regular Coulomb vacuum, the tangent factor consists of r free Abelian vector multiplets and any residual factor is electromagnetically neutral. For a smooth Higgs vacuum, the tangent factor consists of free hypermultiplets; residual vectors require a mixed-branch or decoupled-sector qualification.

Minimal set of statements to remember

1. A branch coordinate is a specified protected scalar primary, not an entire multiplet.
2. A pure vacuum makes VEVs multiplicative only after protected position independence and cluster decomposition are used.
3. Ordinary complex vacua are characters, equivalently maximal ideals under finite generation; Spec contains additional geometric bookkeeping.
4. Coulomb geometry as an affine variety is not yet Seiberg–Witten geometry. The latter needs the low-energy charge lattice and central-charge section.
5. a^I and $a_{D,I}$ are continuous period values; n_e^I and $n_{m,I}$ are the integer charges.
6. Smooth branch geometry controls tangent moduli, not necessarily every massless degree of freedom.

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